

University of Cape Town
Department of Mathematics and Applied Mathematics

2IA Class Test 1
22nd August, 2023
Time: 1,5 hours
Total marks: 105
Full marks: 100

- Read the instructions carefully.
- You have 1,5hrs to complete the test.
- Make sure that no pages or questions are missing.
- Show all your work or reasoning. Answers without work or reasoning shown will receive no mark.

Surname: _____

First name: _____

Student number: _____

Signature: _____

Do not write below this line:

Question:	1	2	3	4	Total
Marks:	10	44	36	15	105
Score:					

1. 10 marks Use the Principle of Mathematical Induction to prove that $n^3 + 2n$ is a multiple of 3 for all $n \geq 1$.

Solution: For $n \geq 1$, consider the statement:

$$p_n: \quad n^3 + 2n \text{ is a multiple of } 3.$$

Let us first prove that p_1 is true:

$$1^3 + 2 \cdot 1 = 1 + 2 = 3 = 3 \cdot 1$$

Next assume that p_k is true for some $k \geq 1$, and let us show that p_{k+1} is also true. The hypothesis induction tells us that

$$k^3 + 2k = 3m, \text{ for some integer } m.$$

From here, we obtain that

$$\begin{aligned} (k+1)^3 + 2(k+1) &= (k^3 + 3k^2 + 3k + 1) + 2k + 2 = \\ &= (k^3 + 2k) + 3k^2 + 3k + 3 = 3m + 3k^2 + 3k + 3 = 3(m + k^2 + k + 1), \end{aligned}$$

which is a multiple of 3. So p_{k+1} holds and the Principle of Mathematical Induction finishes the proof.

2. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**

- (a) 6 marks Suppose that n and m are two integers not both zero, and $d > 1$ is an integer satisfying that $d = xm + yn$ for some integers x and y . Then $d = \gcd(n, m)$.
- (b) 6 marks If $d = \gcd(n, m)$ and $n = da$ and $m = db$. Then $\gcd(a, b) = 1$.
- (c) 6 marks $4^2 - 3 \equiv 1 \pmod{15}$.
- (d) 6 marks If $a \equiv b \pmod{n}$ and $m|n$, then $a \equiv b \pmod{m}$.
- (e) 7 marks If $p > 3$ is a prime, then either $\bar{p} = \bar{1}$ or $\bar{p} = \bar{5}$ in $\mathbb{Z}_6$.
- (f) 7 marks If $\alpha, \beta \in S_n$ commute, then α and β must be disjoint.
- (g) 6 marks If σ fixes k , then σ^{-1} also fixes k .

Solution: (a) **False:** For example: $12 = 1 \cdot 4 + 1 \cdot 8$, and 12 is not the gcd of 4 and 8, since $\gcd(4, 8) = 4$.

(b) **True:** By Bézout's Lemma we can find integers x and y such that $d = xn + ym$. Now, we are assuming that $n = da$ and $m = db$, so

$$d = xn + ym = x(da) + y(db) = d(xa + yb) \Rightarrow d(1 - (xa + yb)) = 0 \Rightarrow xa + yb = 1.$$

This implies that $\gcd(a, b) = 1$.

(c) **True:** $(47 + 3) - 0 = 50$ and $5|50$.

(d) **True:** Assume that $a \equiv b \pmod{n}$ and that $m|n$. Then $n|(a - b)$ and since $m|n$, we get that $m|(a - b)$, which means that $a \equiv b \pmod{m}$.

(e) **True:** Let us begin by noticing that if $\bar{p} = \bar{0}$, then $p \equiv 0 \pmod{6}$; that is, $6|p$, which contradicts the fact that p is prime. If $\bar{p} = \bar{2}$ (respectively, $\bar{p} = \bar{4}$), then $p \equiv 2 \pmod{6}$ (respectively, $p \equiv 4 \pmod{6}$), then $p = 2 + 6k$ (respectively, $p = 4 + 6k$) for some $k \geq 1$. In any case, we get that $2|p$, which is impossible because we are assuming that p is a prime strictly bigger than 3. Lastly, if $p \equiv 3 \pmod{6}$, then $p = 3 + 6k$ for some $k \geq 1$; which yields $3|p$, a contradiction. Thus, the only possibilities are either $\bar{p} = \bar{1}$ or $\bar{p} = \bar{5}$.

Alternatively, notice that since $p > 3$, we get that $\gcd(p, 6) = 1$. This implies that $\bar{p}$ has an inverse in $\mathbb{Z}_6$. The only elements of $\mathbb{Z}_6$ that have a (multiplicative) inverse are $\bar{1}$ or $\bar{5}$. So, either $\bar{p} = \bar{1}$ or $\bar{p} = \bar{5}$.

(f) **False:** Let $n = 3$ and take $\alpha = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ and $\beta = \alpha^2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$. Then $\alpha\beta = \alpha^3 = \varepsilon = \beta\alpha$, and notice that α and β are not disjoint.

(g) **True:** Suppose that σ fixes k , that is, $\sigma(k) = k$. Applying σ^{-1} to both sides of this equality we get that $k = \sigma^{-1}(\sigma(k)) = \sigma^{-1}(k)$, that is, σ^{-1} fixes k .

3. (a) 5 marks Complete the missing parts (marked with a $\star$) in the statement of the **Classical Chinese Remainder Theorem**.

CLASSICAL CHINESE REMAINDER THEOREM. Let m, n, s, t be integers with $m \geq 2$ and $n \geq 2$. Assume that m and n are $(\star)$. Then the simultaneous congruences:

$$x \equiv s \pmod{m}$$

$$x \equiv t \pmod{n}$$

have a common solution. Moreover, any two solutions are congruent modulo $(\star)$.

- (b) Consider the following incomplete proof of the **Classical Chinese Remainder Theorem**:

First, notice that every solution of the congruence equation $x \equiv s \pmod{m}$ has the form (A). Thus, we need to show that the congruence equation (B) has a solution. But this holds since (C). We have proved that the congruence equations $x \equiv s \pmod{m}$ and $x \equiv t \pmod{n}$ have a common solution.

Suppose that both x and y are two common solutions. Then:

$$(D)$$

which imply that both m and n divide $x - y$. The result now follows from the previous lemma.

With reference to this partial proof:

- i. 4 marks Complete the sentence at (A).
 - ii. 4 marks Complete the sentences at (B).
 - iii. 3 marks Complete the sentence at (C).
 - iv. 10 marks Fill in the necessary steps at (D) to conclude that both m and n divide $x - y$.
- (c) 10 marks Find all the solutions to the simultaneous congruences:

$$3x \equiv 2 \pmod{5} \quad \text{and} \quad x \equiv 2 \pmod{3}.$$

Solution: (a) CLASSICAL CHINESE REMAINDER THEOREM. Let m, n, s, t be integers with $m \geq 2$ and $n \geq 2$. Assume that m and n are **coprime**. Then the simultaneous congruences:

$$\begin{aligned} x &\equiv s \pmod{m} \\ x &\equiv t \pmod{n} \end{aligned}$$

have a common solution. Moreover, any two solutions are congruent modulo **mn** .

(b)

i. $x = s + mk$, for some $k \in \mathbb{Z}$.

ii. $s + mk \equiv t \pmod{n}$, or equivalently, $mk \equiv t - s \pmod{n}$

iii. $\gcd(m, n) = 1$

iv. using that congruence is an equivalence relation we obtain

$$\begin{aligned} x &\equiv s \pmod{m} \text{ and } y \equiv s \pmod{m} &\Rightarrow x &\equiv y \pmod{m}, \\ x &\equiv t \pmod{n} \text{ and } y \equiv t \pmod{n} &\Rightarrow x &\equiv y \pmod{n}, \end{aligned}$$

(c) Proceeding like in the proof of the Classical Chinese Remainder Theorem, we first look at the solutions of $x \equiv 2 \pmod{3}$, which are of the form $x = 2 + 3k$, for $k \in \mathbb{Z}$. Substituting in the first congruence we obtain:

$$3(2 + 3k) \equiv 2 \pmod{5} \Leftrightarrow 9k \equiv -4 \pmod{5}$$

Next, since $9 \equiv 4 \pmod{5}$ and $-4 \equiv 1 \pmod{5}$ our equation becomes:

$$4k \equiv 1 \pmod{5}$$

To do so, just notice that $16 \equiv 1 \pmod{5}$, that is, $\bar{4}^{-1} = \bar{4}$ in $\mathbb{Z}_5$. Using this we get that $k \equiv 4 \pmod{5}$. Thus, a common solution of the simultaneous congruences is $x = 2 + 3 \cdot 4 = 14$, and by the Chinese Theorem Remainder we can conclude that all the solutions are:

$$14 + (3 \cdot 5)n = 14 + 15n, \quad n \in \mathbb{Z}.$$

4. (a) 5 marks State Bézout's Lemma.
- (b) Let a, b, n, d be integers such that $n \geq 2$ and $d = \gcd(a, n)$. Prove the following assertions:
- i. 4 marks If the equation $ax \equiv b \pmod{n}$ has a solution, then $d|b$.
 - ii. 6 marks If $d|b$, then the equation $ax \equiv b \pmod{n}$ has a solution.
- Hint:** Use Bézout's Lemma in (b) ii.

Solution: (a) Bézout's Lemma Let m, n be two integers not both zero. Then $d = \gcd(m, n)$ exists and $d = xm + yn$ for some integers x and y .

(b) Let a, b, n, d be integers such that $n \geq 2$ and $d = \gcd(a, n)$.

i. Suppose that the equation $ax \equiv b \pmod{n}$ has a solution, say x_0 . Then $ax_0 \equiv b \pmod{n}$, and so

$$n|(ax_0 - b) \Rightarrow ax_0 - b = kn,$$

for some integer k . But then $b = ax_0 + (-k)n$, and since d is a common divisor of a and n (it is indeed their gcd), we have that d divides any linear combination of x_0 and n ; in particular $d|(ax_0 + (-k)n) = b$, and we are done.

ii. Assume that $d|b$, that is to say that $b = dc$ for some integer c . On the other hand, from $d = \gcd(a, n)$, an application of Bézout's Lemma tells us that

$$d = ua + vn,$$

for some integers u and v .

We claim that $x_0 = uc$ is a solution of the equation $ax \equiv b \pmod{n}$. In fact, we need to check that $ax_0 \equiv b \pmod{n}$; to do so, we need to show that $n|(ax_0 - b)$.

$$ax_0 - b = a(uc) - b$$

On the other hand, from $b = cd = c(ua + vn) = (cu)a + (cv)n$, we get that

$$(cu)a = b - (cv)n.$$

Substituting in the previous equation we obtain

$$ax_0 - b = a(uc) - b = (b - (cv)n) - b = (-cv)n,$$

which shows that $n|(ax_0 - b)$, as desired.